

Migration on a network

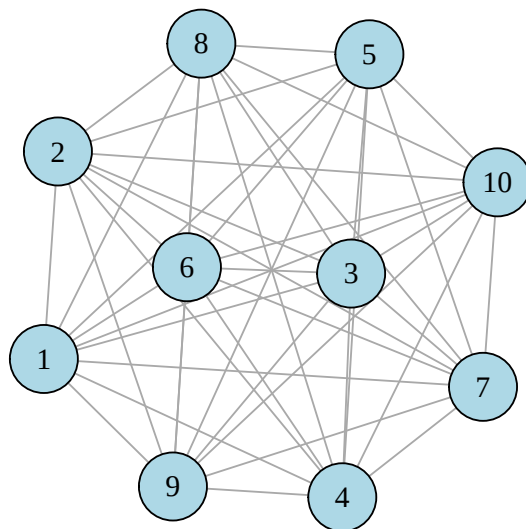
Introduction

Tigers have declined in numbers partly due to loss of habitat connectivity. Here, we use movement along a dynamic network to assess what habitat and connectivity characteristics facilitates or minimizes local extinctions.

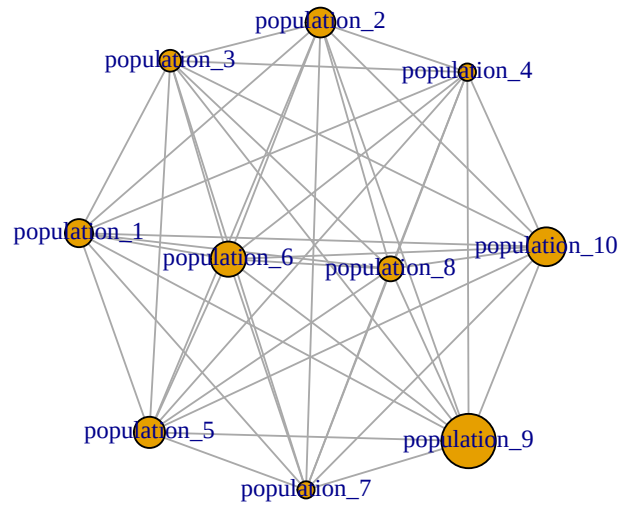
Simple network

- generate a simple network with 10 vertices fully connected.
- assign population sizes of animals to each vertex
- migration is possible through the network
- migration is dependent on population size
- add edge and vertex characteristics
 - Vertex can be vegetated, have food etc, have human habitaion
 - Edges can have weights associated with them due to roadways, forested networks, rivers/mountains, etc

Fully Connected Migration Network



Add vertex characteristics



Add Edge characteristics

